# Probability with Counting

Name _____

When every outcome is equally likely, count the outcomes with permutations or combinations.

$$P(\text{event}) = \frac{\text{number of ways the event can happen}}{\text{total number of possible outcomes}}$$ Count both with the same method.

**Example 1: choosing a group.** A teacher randomly picks 2 students from 5 girls and 4 boys. What is $P$(both girls)?

Total ways to pick 2 of 9: ${}_9C_2 = 36$

Ways to pick 2 of the 5 girls: ${}_5C_2 = 10$

Probability: $P(\text{both girls}) = \frac{10}{36} = \frac{5}{18} \approx 0.28$

**Example 2: a lottery.** Pick 3 different numbers from 1 to 20. Order does not matter. What is $P$(win)?

Total: ${}_{20}C_3 = 1{,}140$. Only 1 of them wins. $P(\text{win}) = \frac{1}{1{,}140} \approx 0.0009$

**Example 3: jobs.** 10 students are in a club. A president and a vice president are chosen at random. What is the probability that Ana is president and Ben is vice president?

Order matters (different jobs). Total: ${}_{10}P_2 = 90$. Only 1 way works. $P = \frac{1}{90}$

**Steps**

1. Decide: does order matter? (permutation or combination)
2. Count all the possible outcomes.
3. Count the outcomes you want.
4. Divide.

Work with a partner. Compare answers when you finish each problem.

1. A committee of 3 is chosen at random from 6 juniors (11th grade) and 4 seniors (12th grade).

   (a) How many committees are possible?

   (b) Find $P$(all seniors).

   (c) Find $P$(all juniors).

2. A lottery: pick 4 different numbers from 1 to 30. Order does not matter. What is the probability of winning with one ticket?

3. 6 runners are in a race. Each order of finish is equally likely. What is the probability that Maya finishes 1st and Leo finishes 2nd?

4. Two cards are dealt from a deck of 52. What is $P$(both aces)? (There are 4 aces.)

5. In a taste test, there are 4 drinks from 4 different brands. A person tastes each drink and matches it to a brand. Each brand is used once. If the person just guesses, what is the probability that all 4 are right?

6. You grab 2 shirts at random from a drawer of 8 shirts. 3 of the shirts are blue. What is $P$(both blue)?

7. A big lottery asks you to pick 5 numbers from 1 to 70 and 1 extra number from 1 to 25.

   (a) Use a calculator. How many different tickets are possible?

   (b) What is the probability of winning with one ticket? Should you expect to win? Explain.